

Practical exercises

— multiplying and dividing by 0.1 and 0.01, and rounding

A Cambridge insert: the decimal arithmetic that every later measurement depends on.

LESSON 7–8

2 × 40 MIN

MATEMATIKA 7, AMALIY MASHQLAR

S8 3.1–3.2

LESSON CLOCK

15'

25'

25'

20'

5'

Moving the point	15 min
Dividing by a small number	25 min
Rounding	25 min
Estimating	20 min
Homework	5 min

LEARNING OBJECTIVES

- Multiply and divide by 0.1, 0.01 and 0.001 by moving the decimal point.
- Explain why dividing by 0.1 makes a number larger.
- Round to a given number of decimal places and of significant figures.
- Estimate an answer before computing it.

TEXTBOOK

National	pp. 22–25
Cambridge	Stage 8, pp. 26–34
Workbook	Exercise 3.1–3.2

01

Explanation

Moving the point

OPERATION	EFFECT	EXAMPLE
$\times 10$	point moves 1 right	$3.7 \rightarrow 37$
$\times 0.1$	point moves 1 left	$3.7 \rightarrow 0.37$
$\times 0.01$	point moves 2 left	$3.7 \rightarrow 0.037$
$\div 0.1$	point moves 1 right	$3.7 \rightarrow 37$
$\div 0.01$	point moves 2 right	$3.7 \rightarrow 370$

Multiplying by 0.1 is the same as dividing by 10; dividing by 0.1 is the same as multiplying by 10.

$0.1 = \frac{1}{10}$, AND THAT EXPLAINS EVERYTHING

Dividing by $\frac{1}{10}$ asks “how many tenths are there in it?” — and there are ten times as many tenths as there are wholes.

Dividing by a small number

The result of dividing by a number less than 1 is **larger** than what you started with. This surprises almost everyone at first.

QUESTION	IN WORDS	ANSWER
$6 \div 2$	how many 2s in 6?	3
$6 \div 1$	how many 1s?	6
$6 \div 0.5$	how many halves?	12
$6 \div 0.1$	how many tenths?	60
$6 \div 0.01$	how many hundredths?	600

“DIVIDING MAKES IT SMALLER” IS FALSE

It is true only when dividing by a number greater than 1. This is one of the most persistent wrong beliefs in school arithmetic, and reading the division as “how many ... in ...?” cures it.

Rounding

RULE	STATEMENT
look at	the first digit you are discarding
if it is 0–4	leave the kept digit alone
if it is 5–9	add 1 to the kept digit

Decimal places count from the point; **significant figures** count from the first non-zero digit.

NUMBER	2 D.P.	2 S.F.
3.4567	3.46	3.5
0.004829	0.00	0.0048
1954	1954.00	2000

LEADING ZEROS ARE NOT SIGNIFICANT

In *0.004829* the first significant figure is the 4. That is why the two columns above differ so sharply for that row.

Estimating

Round every number to *1* significant figure, compute, and compare with the exact answer.

CALCULATION	ESTIMATE	EXACT
4.8×21.3	$5 \times 20 = 100$	102.24
$39.6 \div 0.21$	$40 \div 0.2 = 200$	188.6
6.9×0.48	$7 \times 0.5 = 3.5$	3.312

ESTIMATE FIRST, THEN COMPUTE

An estimate that is ten times out from the answer means a decimal point was misplaced. Thirty seconds of estimating catches errors no amount of re-checking the digits will find.

02 Worked examples**EXAMPLE 1** Compute 4.7×0.01 and $4.7 \div 0.01$.

① $\times 0.01$ moves the point 2 places left.

② $4.7 \times 0.01 = 0.047$

③ $\div 0.01$ moves it 2 places right.

④ $4.7 \div 0.01 = 470$

ANSWER 0.047 and 470 **EXAMPLE 2** Round 3.4567 to two decimal places and to two significant figures.

① 2 d.p.: keep 3.45, next digit 6.

② ≈ 3.46

③ 2 s.f.: keep 3.4, next digit 5.

④ ≈ 3.5

ANSWER 3.46 and 3.5

EXAMPLE 3 Estimate $39.6 \div 0.21$, then say whether *18.86* is a plausible answer.

- ① $40 \div 0.2$
One significant figure each.

- ② $= 200$

- ③ *18.86* is ten times too small.

- ④ The true answer is about *189*.
A misplaced point.

ANSWER

No — about *189*

03 Interactive model

Ask how many 10-tiyin coins make a sum before doing any division by 0.1; the arithmetic follows from the question, not from a rule.

This model could not start: den is not a function

04 Terminology

Write these on the board at the start. Learners meet the national programme in Uzbek or Russian and the Cambridge papers in English — the three columns are the same idea.

ENGLISH	O'ZBEKCHA	РУССКИЙ
Decimal point	O'nli nuqta	Десятичная запятая
Decimal place	O'nli xona	Десятичный знак
Significant figure	Ma'noli raqam	Значащая цифра
To round	Yaxlitlash	Округлить
Estimate	Baholash	Оценка
Tenth	O'ndan bir	Десятая
Hundredth	Yuzdan bir	Сотая
Approximation	Taqribiy qiymat	Приближение

Stress the words in bold in the explanation above; ask for the Uzbek and Russian back before moving on.

05 Quick check**Check yourself**

$\times 0.1$ moves the point:

one right

one left

two right

two left

$\div 0.01$ moves the point:

two left

two right

one right

not at all

$6 \div 0.1$ equals:

0.6

6

60

0.06

Dividing by a number less than 1:

makes it smaller

makes it larger

leaves it

is impossible

3.4567 to 2 s.f.:

3.45

3.46

3.5

3.4

In 0.00482 the first significant figure is:

0

4

8

2

06 Practice · 21 problems

Seven at each level. Set the easy row for everyone, the medium row for the main body of the class, and the hard row for those who finish early.

Easy · warm the idea up

7 PROBLEMS

1. 3.7×0.1

ANSWER 0.37

2. 3.7×0.01

ANSWER 0.037

3. $3.7 \div 0.1$

ANSWER 37

4. $3.7 \div 0.01$

ANSWER 370

5. $6 \div 0.5$

ANSWER 12

6. $6 \div 0.1$

ANSWER 60

7. 4.7×0.01

ANSWER 0.047

Medium · the standard question

7 PROBLEMS

1. 3.4567 to 2 d.p.

ANSWER 3.46

2. 3.4567 to 2 s.f.

ANSWER 3.5

3. 0.004829 to 2 s.f.

ANSWER 0.0048

4. 1954 to 2 s.f.

ANSWER 2000

5. Estimate 4.8×21.3

ANSWER 100

6. Estimate $39.6 \div 0.21$

ANSWER 200

7. $0.42 \div 0.01$

ANSWER 42

Hard · stretch and reason

7 PROBLEMS

1. $0.6 \times 0.4 \div 0.02$

ANSWER 12

2. $(0.3)^2 \div 0.09$

ANSWER 1

3. Estimate $\frac{58.7 \times 0.51}{0.19}$

ANSWER ≈ 150

4. 12.345 to 3 s.f.

ANSWER 12.3

5. 0.09995 to 3 d.p.

ANSWER 0.100

6. A rope of 8.4 m cut into 0.35 m pieces

ANSWER 24 pieces

7. Why does $\div 0.1$ multiply by 10?

ANSWER There are ten tenths in one whole

07 Homework**Homework — 5 tasks**

Estimate every calculation to one significant figure before doing it exactly.

1. Compute 5.6×0.1 , 5.6×0.001 , $5.6 \div 0.1$ and $5.6 \div 0.001$.
2. Round 27.4863 to 1, 2 and 3 decimal places.
3. Round 0.0035729 to 2 and 3 significant figures.
4. Estimate 7.8×0.49 and then compute it exactly.

5. A ribbon of 6.3 m is cut into pieces of 0.45 m . How many pieces?